

Q1A no. 1

(a)

RS latch with Nand gates:-

① If $R=0$ and $S=0$:-

This is invalid state

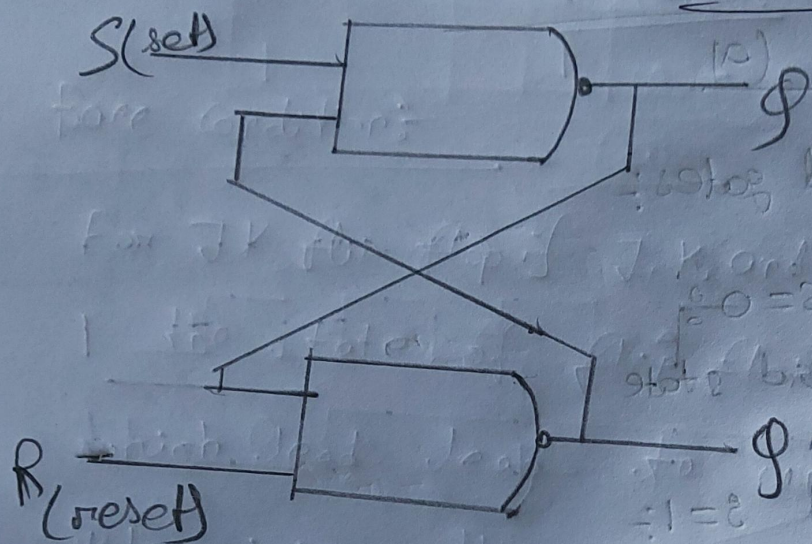
② If $R=0$ and $S=1$:- $Q=1$ (set state) $Q'=0$ ③ If $R=1$ and $S=0$:- $Q=0$ (reset state) $Q'=1$ ④ If $R=1$ and $S=1$ Q and Q' maintain their previous state.

R	S	Q	Q'
0	0	Invalid	Invalid
0	1	1	0
1	0	0	1
1	1	No change	No change

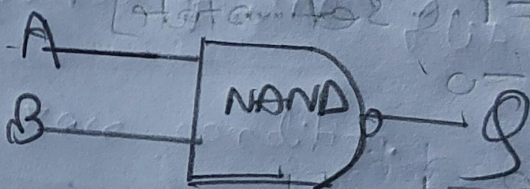
ES-2-1

(b) OR gate

SR latch using NAND



S	R	Q	Q'
1	0	0	1
1	1	0	1
0	1	1	0
1	0	1	0
0	0	1	1



A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

(b)

Available (SR)

Req $\rightarrow$ (D) ffSR to D ff conversionCharacteristics Table:Excitation Table:

S_n	D	S_{n+1}	S	R
0	0	0	0	X
0	1	1	1	0
1	0	0	0	1
1	1	1	X	0

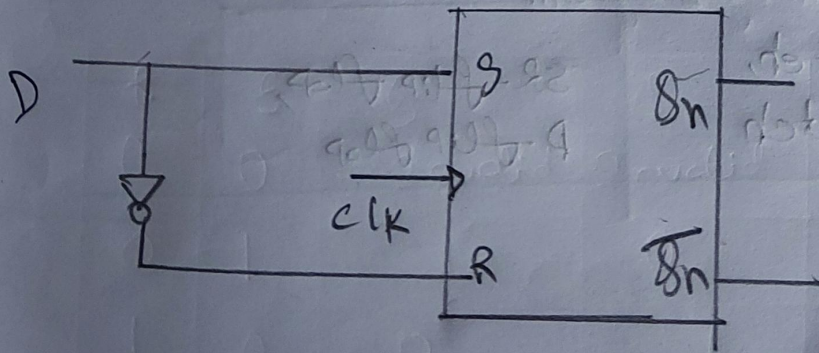
S_n	S_{n+1}	S	R
0	1	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S_n	D	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

$$S = D$$

S_n	D	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

$$R = \bar{D}$$



Q/A.no.2

(a)

Race Condition

For JK flip-flop if J, K and clock are equal to 1 the state of flip flop keeps on toggling which leads to uncertainty in determining the output of the flip flop, This is called Race condition.

Topic	Latch	Flip flop
Sensitivity	Level-sensitive	Edge-sensitive
State Change	Transparent when enabled	At clock edges
Common Types	SR latch, D latch	SR-flip flop, D-flip flop

(b)

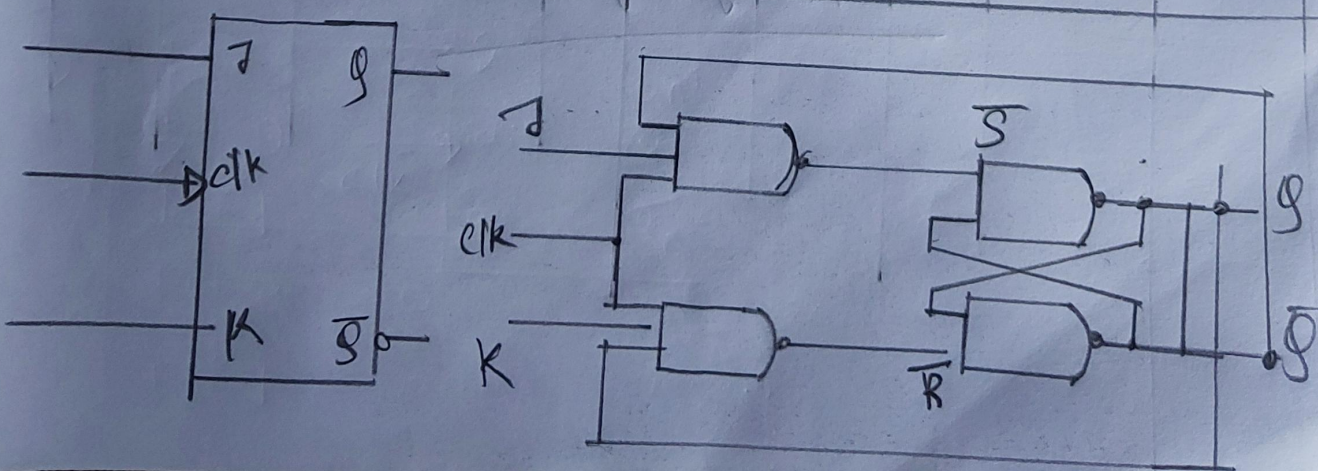
JK ff. eliminates the intermi indeterminate state of the SR flip flop by making both inputs high ($J=1, K=1$) toggle the output instead of causing an undefined state.

JK ff Truth table:-

Characteristics Table

clk	J	K	Q_{t+1}
0	X	X	Q_t
1	0	0	Q_t
1	0	1	0
1	1	0	1
1	1	1	$\bar{Q}_t$

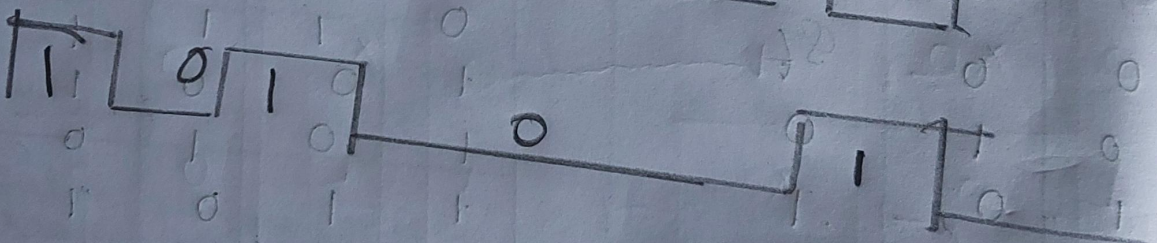
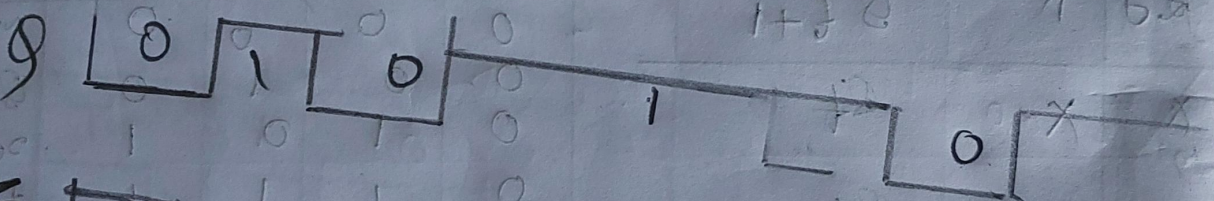
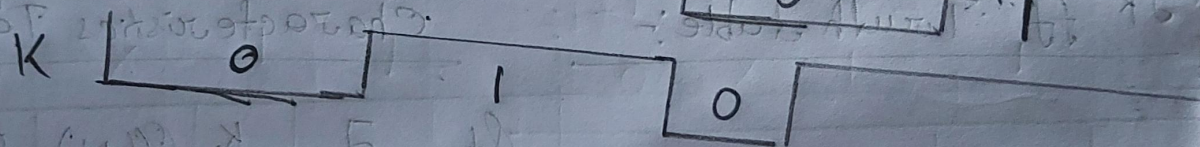
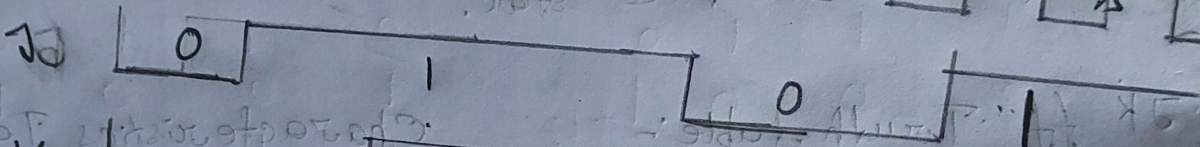
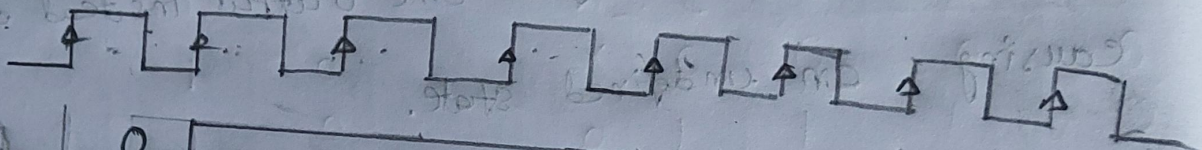
Q_t	J	K	Q_{t+1}	Comt
0	0	0	0	No change
0	0	1	0	Q clear
0	1	0	1	Q set
0	1	1	1	toggle
1	0	0	1	No change
1	0	1	0	Q clear
1	1	0	1	Q set
1	1	1	0	toggle



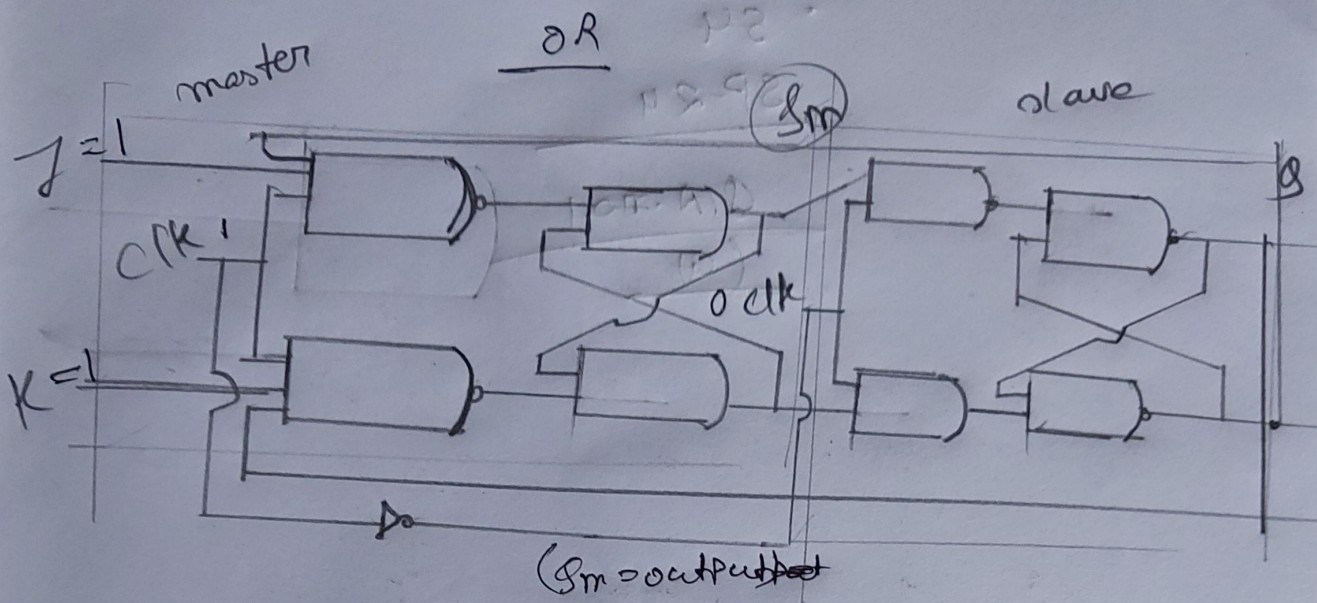
JK

0	0	1	1
1	1	1	1

$$Q_{t+1} = J\bar{Q} + RQ$$



Timing Diagram



$J=K=1$ $clk=1$ $\phi m = \text{Active}$ $Q = \text{memory}$

$J=K=1$ $clk=0$ $\phi m = \text{memory}$ $Q = \text{Active}$

